1.7 Equilibria

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.7 Equilibria
Difficulty	Easy

Time allowed: 40

Score: /30

Percentage: /100

Question 1a

This question is about dynamic equilibria.

State what is meant by the term dynamic equilibrium.

[2 marks]

Question 1b

A general reaction exists in the following dynamic equilibrium.

$$3A + 2B \rightleftharpoons C_2 + D$$

Which part of the equation shows that this reaction is a dynamic equilibrium?

[1 mark]

Question 1c

The gases nitrogen dioxide, NO_2 (g), and nitrogen tetroxide, N_2O_4 (g), exist in the following dynamic equilibrium.

$$2NO_2(g) \rightleftharpoons N_2O_4(g)$$

An increase in pressure shifts the equilibrium to the product's side and increases the yield of nitrogen tetroxide.

State why the increase in pressure shifts the equilibrium to the product's side.

[1 mark]

Question 2a

During the manufacture of sulfuric acid in the Contact process, sulfur dioxide, SO_2, is oxidised into sulfur trioxide, SO_3.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \quad \Delta H = -197 \text{ kJ mol}^{-1}$$

State the effect on the rate of reaction when the temperature increases.

[1 mark]

Question 2b

State, with a reason, the effect on the yield of sulfur trioxide when the temperature increases.

[2 marks]

Question 2c

Describe and explain the effect of decreasing the pressure on the yield of sulfur trioxide.

[3 marks]

Question 3a

Ammonia is manufactured from nitrogen and hydrogen using the Haber process.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

State the effect that the addition of an iron catalyst would have on the position of the equilibrium.

[1 mark]

Question 3b

Write the expression for K_c for the production of ammonia.

[1 mark]

Question 3c

At 450 °C, the equilibrium concentrations are N_2 (g): 17.4 mol dm^{-3}, H_2 (g): 0.85 mol dm^{-3}, NH_3 (g): 1.25 mol dm^{-3}.

Calculate the value of K_c to three significant figures.

[2 marks]

Question 4a

Dinitrogen pentoxide, N_2O_5 (g), decomposes into oxygen and nitrogen dioxide via the following equation.

$$2N_2O_5\,(g) \rightleftharpoons O_2\,(g) + 4NO_2\,(g)$$

Give the expression for the equilibrium constant, K_p, for this equilibrium.

K_p =

[1 mark]

Question 4b

The partial pressures at temperature *T* are shown in Table 4.1

Table 4.1

	N_2O_5 (g)	O_2 (g)	NO_2 (g)
Partial pressure (Pa)	10 000	400	600

i)

Use your answer to part **(a)** and Table 4.1 to calculate a value for K_p at temperature *T*.

K_p = ...

[2]

ii)

Determine the units for K_p.

Units = ...

[1]

[3 marks]

Question 4c

The reaction is repeated at a lower pressure.

State the effect of this on K_p.

[1 mark]

Question 5a

Define a Brønsted–Lowry acid.

[1 mark]

Question 5b

Which species in the following reaction acts as a Brønsted-Lowry base.

$$HSO_4^-(aq) + H_3O^+(aq) \rightleftharpoons H_2SO_4(aq) + H_2O(l)$$

[1 mark]

Question 5c

Which species in the following equation is acting as a Brønsted-Lowry acid.

$$CO_3^{2-}(aq) + H^+(aq) \rightleftharpoons HCO_3^-(aq)$$

[1 mark]

Question 5d

Propanoic acid, $CH_3CH_2CO_2H$ is classed as a weak acid.

Write an equation to show the dissociation of propanoic acid.

[1 mark]

Question 6a

A pH curve can be drawn from the data recorded from the following procedure

Step **1** Placing a fixed volume of a HCl (aq) into a beaker

Step **2** Add KOH (aq) in known small portions from a burette and stir

Step **3** Use pH meter to record pH after every addition of alkali

Sketch a curve on the graph in Fig. 6.1 for a strong acid - strong base titration that could be drawn after this procedure is carried out.

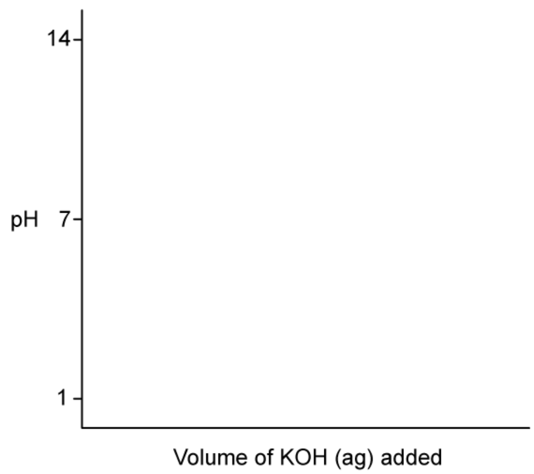


Fig. 6.1

[3 marks]

Question 6b

A titration was performed between ethanoic acid, CH_3CO_2H (aq) and sodium hydroxide, NaOH (aq). Using the information in Table 6.1, suggest which indicator would be suitable for this titration.

Table 6.1

Indicator	pH range
Pentamethoxy red	1.2 - 3.2
Naphthyl red	3.7 - 5.0
4–nitrophenol	5.6 - 7.0
Cresol purple	7.6 - 9.2

i)
Write an equation for the reaction between ethanoic acid and sodium hydroxide

[1]

ii)
Using the information in Table 6.1, suggest which indicator would be suitable for this titration

[1]

[2 marks]

Question 6c

Apart from using an indicator or pH probe, describe **one** test that could be used to show that hydrochloric acid is a stronger acid than ethanoic acid of the same concentration.

[2 marks]

Model Answers: Easy

1a

a) A dynamic equilibrium is:

- A reaction where the rate of the forward reaction is the same as the rate of the backward reaction (in a closed system); [1 mark]
- A reaction where the concentrations of the reactants and products are constant; [1 mark]
  Do not accept where the concentrations of the reactants and products are the same

[Total: 2 marks]

- The second mark about the concentrations is often answered incorrectly
- Be careful to state that the concentration of reactants and products **remains constant** as you will lose a mark if you do not write this
  - Once a reversible reaction reaches dynamic equilibrium, the concentration of the reactants and products does not change
  - This is why you must say that they remain constant and **NOT** that they are the same

1b

b) The part of the equation that shows this reaction is a dynamic equilibrium is:

- $\rightleftharpoons$ / the reversible arrow; [1 mark]

[Total: 1 mark]

- A dynamic equilibrium is a reversible reaction where the reactants form products but the products also form reactants
- This means that the equation should have a $\rightleftharpoons$ arrow

1c

c) The increase in pressure shifts the equilibrium to the product's side because:

- There are more moles / molecules on the reactant's side
  OR
  There are less moles / molecules on the product's side
  OR
  There are 2 moles / molecules of reactant but only 1 mole / molecule of product; [1 mark]

[Total: 1 mark]

- **Careful:** Generally, exam answers about pressure changes in equilibria reactions lose marks because they are not specific enough
  - Talking about the general idea of Le Chatelier's principle and how increasing the pressure favours the side with the fewest moles / molecules will not gain the marks
  - You must say which side has the fewest moles / molecules
- This is why mark schemes often say about ignoring references to Le Chatelier's principle because the examiner wants to see you apply it to the specific reaction that you have been given

2a

a) When the temperature increases, the rate of reaction will:

- Increase; [1 mark]

[Total: 1 mark]

- **Careful:** This question is about the rate of reaction
- Increasing the temperature will increase the rate of the forwards and backwards reactions
- This happens because the particles gain energy and collide more often, resulting in more successful collisions

2b

b) When the temperature increases, the yield will:

- Decrease; [1 mark]

- (Because,) the forward reaction is exothermic
  OR
  (Because,) the backward reaction is endothermic; [1 mark]

[Total: 2 marks]

- **Careful:** This question is about product yield
- Examiners often comment on questions about yield linked to changes in temperature
- Students typically make one of two mistakes with these questions:
  - Over-answering
    - The best correct answer (the forward reaction is exothermic) to this question is simple
    - Often answers will start trying to incorporate ideas about collision theory (particle's energy, frequency of collisions, success of collisions) which presumably comes from GCSE where answers involving changes in temperature are multi-step / more complicated
  - Under-answering
    - This is most commonly failing to say which reaction is exothermic / endothermic

2c

c) Decreasing the pressure:

- Decreases the yield; [1 mark]

Explanation:

- The equilibrium shifts to the left; [1 mark]
- (Because,) the left-hand side has more (gaseous) moles / molecules
  OR
  (Because,) the right-hand side has less (gaseous) moles / molecules; [1 mark]

[Total: 3 marks]

- **Careful:** This question is about decreasing pressure and yield
- This means that your answer should be talking about shifts in equilibrium
- You should score well on this question if you:
  - State what happens to the yield

- State which side the equilibrium moves to
- Specifically state why the equilibrium shifts - this means saying which side has more moles / molecules or comparing the number of moles / molecules on each side

- Be aware, that examiners will sometimes ask you questions where the change in conditions causes the rate to increase and the yield to decrease or vice versa

3a

a) The addition of an iron catalyst would have:

- No effect / do nothing; [1 mark]

[Total: 1 mark]

- Catalysts increase the rate of both the forwards and backwards reaction equally
- This means that they only help a reaction to reach equilibrium faster and have no effect on the position of the equilibrium once this is reached

3b

b) The expression for K_c for the production of ammonia is:

- $K_c = \frac{[NH_3]^2}{[N_2]\,[H_2]^3}$; [1 mark]

[Total: 1 mark]

- **Remember:** $K_c = \frac{\text{[products]}}{\text{[reactants]}}$ and the concentrations are raised to a power that is equal to the stoichiometric coefficient of each chemical
  - There are 2 ammonia moles / molecules produced so this creates the $[NH_3]^2$ term in the K_c expression
  - There is 1 reactant nitrogen mole / molecule which creates the $[N_2]$ term in the K_c expression
  - There are 3 reactant moles / molecules of hydrogen which gives the $[H_2]^3$ term in the K_c expression
- Placing each of these terms appropriately in the K_c expression gives:
  - $K_c = \frac{[NH_3]^2}{[N_2]\,[H_2]^3}$

3c

c) The value of K_c to three significant figures is:

- $K_c = \frac{[NH_3]^2}{[N_2]\,[H_2]^3} = \frac{[1.25]^2}{[17.4]\,[0.85]^3}$; [1 mark]
- $K_c = 0.146$; [1 mark]

[Total: 2 marks]

- You are given the values that you require for this calculation
    - So, you just need to substitute them into the equation and complete the equation

4a

a) The K_p expression is:

The K_p expression is:

- $K_p = \frac{p_{O_2} \times (p_{NO_2})^4}{(p_{N_2O_5})^2}$

  OR
- $K_p = \frac{p_{O_2} \times p^4{}_{NO_2}}{p^2{}_{N_2O_5}}$; [1 mark]

[Total: 1 mark]

- Take your time writing these equations and double check your answer
  1. Have you included the sign for partial pressure, p?
  2. Have you written $K_p = \frac{\text{products}}{\text{reactants}}$?
  3. Have you taken the stoichiometry of each reactant and product into account?

4b

b)

i) To calculate a value for K_p:

- $K_p = \dfrac{400 \times (600)^4}{(10\,000)^2}$; [1 mark]
- K_p = 518 400; [1 mark]

ii) To deduce the units:

- $K_p = \dfrac{Pa^5}{Pa^2} = Pa^3$; [1 mark]

[Total: 3 marks]

- Use your K_p expression to substitute in the numbers given in Table 4.1
- There is no need to work out the partial pressures as they have been given to you
- To determine the units write out the expression containing Pa
- Units = $\dfrac{Pa \times Pa^4}{Pa^2} = Pa^3$

2. Have you written $K_p = \frac{\text{products}}{\text{reactants}}$?
3. Have you taken the stoichiometry of each reactant and product into account?

4b

b)

i) To calculate a value for K_p:

- $K_p = \frac{400 \times (600)^4}{(10\,000)^2}$; [1 mark]
- $K_p = 518\,400$; [1 mark]

ii) To deduce the units:

- $K_p = \frac{Pa^5}{Pa^2} = Pa^3$; [1 mark]

[Total: 3 marks]

- Use your K_p expression to substitute in the numbers given in Table 4.1
- There is no need to work out the partial pressures as they have been given to you
- To determine the units write out the expression containing Pa
- Units = $\frac{Pa \times Pa^4}{Pa^2} = Pa^3$

4c

c) The effect of this on K_p is:

- K_p would not change; [1 mark]

[Total: 1 mark]

- Only temperature changes permanently affect the value of K_p

5a

a) A Brønsted–Lowry acid is:

- (A species which) donates H^+/protons; [1 mark]

[Total: 1 mark]

- A Brønsted–Lowry acid is a proton/ H^+ donor and a Brønsted–Lowry base is a proton/ H^+ acceptor.

5b

b) The species acting as a Brønsted base are:

- HSO_4^-
  AND
  H_2O; [1 mark]

[Total: 1 mark]

- The Brønsted-Lowry definition states that bases are proton acceptors
- From the equation, we can see HSO_4^- accepts a proton to form H_2SO_4
- In the reverse reaction, H_2O also accepts a proton to form H_3O^+
- Be careful as you need to identify both of these for the mark
- You should be able to deduce both Brønsted-Lowry acids and bases in chemical equations
- You can do this by looking where the hydrogens move within the equation

5c

c) the species acting as a Brønsted-Lowry acid is:

- HCO_3^-; [1 mark]

[Total: 1 mark]

- The Brønsted-Lowry definition states that acids are proton donors
- From the equation we can see in the reverse reaction HCO_3^- donates a proton to form CO_3^-
- You should be able to deduce both Brønsted-Lowry acids and bases in chemical equations
- You can do this by looking where the hydrogens move within the equation

5d

d) Propanoic dissociates by:

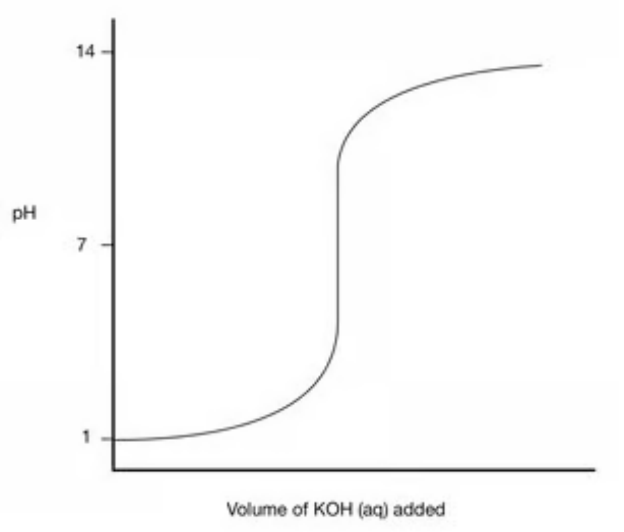
- $CH_3CH_2CO_2H\,(aq) \rightleftharpoons CH_3CH_2COO^-\,(aq) + H^+\,(aq)$
  OR
- $CH_3CH_2CO_2H\,(aq) + H_2O(l) \rightleftharpoons CH_3CH_2COO^-\,(aq) + H_3O^+\,(aq)$; [1 mark]

[Total: 1 mark]

- For weak acids you must use a reversible arrow as they only partially (or incompletely) dissociates in aqueous solution
- The position of equilibrium therefore lies to the left
- Hydrogen ions in aqueous solution are written as H_3O^+ but it is acceptable to simplify its structure to H^+

6a

a) The correct pH curve is:



- Curve starts between pH 0–3; [1 mark]
- Equivalence point at pH 7; [1 mark]
- pH curve ends between pH 12–14; [1 mark]

[Total: 3 marks]

- You are told in the question that both the acid (HCl) and alkali (KOH) are strong
  - The acid will therefore have a pH between 0–3
  - The pH curve will finish between 12–14

- The equivalence point for a strong acid - strong alkali pH curve will be at pH 7
- Make sure pH 7 is halfway up the vertical section of the curve (point of inflection)

6b

b)

i) The equation is:

- CH_3CO_2H (aq) + NaOH (aq) → CH_3CO_2Na (aq) + H_2O (l); [1 mark]

ii) A suitable indicator for a strong base weak acid titration is:

- Cresol purple; [1 mark]

[Total: 2 marks]

- The reaction between ethanoic acid and sodium hydroxide is a neutralisation reaction which will form a salt and water
- **Remember**: A suitable indicator will need to change colour at the equivalence point
- The equivalence point is the point at which just sufficient reagent is added to react completely with a substance
- The point in the titration process which is indicated by colour change of the indicator is called the endpoint
- It is the point where the analyte has **completely** reacted with the titrant
- The equivalence point is **above** pH 7 for a strong base weak acid titration

6c

c) A test to show the difference in acid strength could be:

- Testing the electrical conductivity; [1 mark]
- Hydrochloric acid conducts better than ethanoic acid; [1 mark]

OR

- Adding a reactive metal / carbonate / hydrogen carbonate; [1 mark]
- Hydrochloric acid shows greater effervescence than ethanoic acid; [1 mark]

OR

- Adding sodium hydroxide solution; [1 mark]
- Hydrochloric acid would produce a greater temperature rise than ethanoic acid; [1 mark]

[Total: 2 marks]

- There are many examples you can use, but to get full credit you must mention the technique as well as how the result would set apart hydrochloric acid from ethanoic acid
- If using the last method, be careful not to describe a titration as this would not be a suitable technique since the same volume and same concentration of the two acids would react with an identical volume of sodium hydroxide